

STATS 201A Lecture 3

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1 October 2026

$$Y = \mathbb{X}\beta + \varepsilon$$

$$Z = (\mathbb{P}_1(x), \mathbb{P}_2(x), \mathbb{P}_3(x), \mathbb{P}_4(x))$$

$$Z : 5 \times 4$$

$$Z^{\mathsf{T}}Z = \begin{pmatrix} 10 & & & \\ & 14 & & \\ & & 10 & \\ & & & 70 \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & \frac{\mathbb{P}(x)}{\sqrt{10}} & \frac{\mathbb{P}_2(x)}{\sqrt{14}} & \frac{\mathbb{P}_3(x)}{\sqrt{10}} & \frac{\mathbb{P}_4(x)}{\sqrt{70}} \end{pmatrix}$$

$$\mathbb{X}^{\mathsf{T}}\mathbb{X} = \begin{pmatrix} 5 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \\ & & & & 1 \end{pmatrix}$$

$$\beta = (\mathbb{X}^{\mathsf{T}}\mathbb{X})^{-1} \mathbb{X}^{\mathsf{T}}Y = \begin{pmatrix} \cdot \\ \cdot \end{pmatrix} \mathbb{X}^{\mathsf{T}}Y$$

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbb{X}^{\mathsf{T}}\mathbb{X})^{-1} = \sigma^2 \begin{pmatrix} 5 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \\ & & & & 1 \end{pmatrix}^{-1}$$

$$\text{Var}(\hat{\beta}_0) = \frac{\sigma^2}{25}$$

$$\text{Var}(\hat{\beta}_i) = \frac{\sigma^2}{5}, \quad i = 1, 2, 3, 4$$

$$Y_{ij} = \eta + \tau_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$$

$$\hat{y}_{ij} = \bar{y}_i \quad r_{ij} = y_{ij} - \hat{y}_{ij}$$